

Netflix Inspired AI Movie Recommendation System

DecodeLabs Internship

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Abstract

The Netflix-Inspired AI Movie Recommendation System is a web-based application developed to help users discover movies based on their interests and genre preferences. The system analyzes the genres selected by the user and recommends relevant movies using a similarity-based scoring approach. Each recommendation is assigned a match score, allowing users to understand how closely a movie aligns with their preferences.

The application is built using React.js and Vite and features a modern Netflix-inspired user interface. Additional functionalities such as movie search, recommendation ranking, movie posters, and best-match highlighting enhance the overall user experience. This project demonstrates the practical implementation of recommendation system concepts while strengthening frontend development and problem-solving skills.

Introduction

With the rapid growth of online streaming platforms, users are presented with thousands of entertainment options, making content discovery increasingly difficult. Recommendation systems help solve this challenge by suggesting relevant content based on user preferences and interests. Popular platforms such as Netflix, YouTube, and Spotify extensively use recommendation engines to improve user engagement and satisfaction.

The Netflix-Inspired AI Movie Recommendation System is designed to explore the fundamentals of recommendation systems through a genre-based approach. Users can select their preferred genres and receive personalized movie suggestions ranked according to similarity scores. The project combines recommendation logic with a modern and interactive user interface, providing

an engaging movie discovery experience while demonstrating key concepts of web development and recommendation algorithms.

Objectives

The main objective of this project is to develop an intelligent movie recommendation platform that assists users in finding movies matching their interests. The system aims to simplify content discovery through genre-based recommendations and similarity scoring techniques.

The specific objectives are:

- To build a responsive and user-friendly movie recommendation application.
- To recommend movies based on user-selected genres.
- To implement search and filtering functionality.
- To calculate and display recommendation match scores.
- To create a Netflix-inspired user interface for enhanced user experience.
- To understand and apply the fundamentals of recommendation systems and modern frontend development.

Technology Stack

The Netflix-Inspired AI Movie Recommendation System is developed using modern web development technologies to ensure performance, scalability, and an engaging user experience.

Frontend

React.js was used to build the user interface using reusable components and efficient state management. It enables dynamic rendering and improves the overall user experience.

Build Tool

Vite was used as the development and build tool due to its fast startup time, efficient bundling, and optimized production builds.

Programming Language

JavaScript (ES6+) was used for implementing the recommendation logic, component interactions, and application functionality.

Styling

CSS3 was used to design the Netflix-inspired user interface, including responsive layouts, movie cards, search functionality, and visual effects.

Version Control

Git was used for tracking project changes, managing versions, and maintaining the development workflow.

Repository Hosting

GitHub was used to store the source code, manage project versions, and facilitate collaboration and deployment.

Deployment

Vercel was used to deploy the application and make it accessible through a live web URL with automatic updates from GitHub.

Together, these technologies provide a robust and efficient platform for building and deploying the movie recommendation system.

System Design

The Netflix-Inspired AI Movie Recommendation System follows a simple and efficient architecture designed to provide personalized movie recommendations based on user preferences.

The system begins when a user selects one or more movie genres through the user interface. The recommendation engine then compares the selected genres with the genres associated with each movie in the dataset. Based on the number of matching genres, a similarity score is calculated for every movie.

Movies with higher similarity scores are ranked higher and displayed as recommendations. The system also supports movie search functionality, allowing users to quickly find specific movies from the recommendation list. The final recommendations are presented through a visually

appealing Netflix-inspired interface, complete with movie posters, match percentages, and recommendation explanations.

Workflow

User → Select Genres → Recommendation Engine → Similarity Score Calculation → Ranked Recommendations → Search & Filter → Display Results

Features

The Netflix-Inspired AI Movie Recommendation System provides several features to enhance movie discovery and user experience.

1. **Genre-Based Recommendations:** Suggests movies according to the genres selected by the user.
2. **Similarity Score Calculation:** Calculates and displays a match percentage for each recommended movie.
3. **Movie Search Functionality:** Allows users to search movies by title.
4. **Best Match Highlight:** Identifies and highlights the highest-ranked recommendation.
5. **Movie Posters:** Displays movie posters for better visual representation.
6. **Recommendation Explanation:** Shows why a movie was recommended based on matching genres.
7. **Responsive User Interface:** Provides a seamless experience across different screen sizes.
8. **Netflix-Inspired Design:** Features a modern streaming-platform-inspired interface for improved user engagement.

These features work together to create an interactive and personalized movie recommendation experience.

Recommendation Algorithm

The system uses a genre-based similarity recommendation algorithm to suggest movies. When a user selects one or more genres, the system compares these genres with the genres associated with each movie in the dataset.

A similarity score is calculated based on the number of matching genres. Movies with higher scores are considered more relevant and are ranked higher in the recommendation list. The movie with the highest score is highlighted as the Best Match.

Formula

$$\text{Match Score (\%)} = (\text{Number of Matched Genres} / \text{Number of Selected Genres}) \times 100$$

This approach ensures that users receive recommendations closely aligned with their interests and preferences.

Implementation

The system was developed using React.js and Vite. User-selected genres are processed through a similarity-based recommendation algorithm, which calculates match scores and displays ranked movie recommendations. Additional features such as movie search, posters, and best-match highlighting were integrated to enhance user experience.

Testing

The application was tested for genre selection, recommendation generation, search functionality, score calculation, and responsive design. All features were verified to ensure accurate recommendations and smooth user interaction.

Future Scope

The current system can be further enhanced by incorporating advanced features and technologies to improve recommendation accuracy and user experience. Some possible future enhancements include:

1. Integration with the TMDB API to access real-time movie data and posters.
2. Implementation of User Authentication for personalized experiences.
3. Addition of a Watchlist Feature to save movies for later viewing.
4. Development of User Profiles with individual preferences and viewing history.
5. Integration of Machine Learning Algorithms for more intelligent recommendations.
6. Addition of Movie Ratings and Reviews to improve recommendation quality.

7. Support for Multiple Recommendation Techniques such as collaborative filtering and content-based filtering.
8. Implementation of Dark/Light Theme Switching for better accessibility.
9. Addition of Trending and Popular Movies Sections.
10. Development of a Mobile-Friendly Progressive Web App (PWA) for improved accessibility across devices.

These enhancements would transform the system from a basic recommendation platform into a more intelligent and personalized entertainment application.

Learning Outcomes

Understanding recommendation systems, React architecture, frontend development, deployment, and version control.

Conclusion

The project successfully demonstrates recommendation system fundamentals while providing a professional streaming-platform-inspired experience.